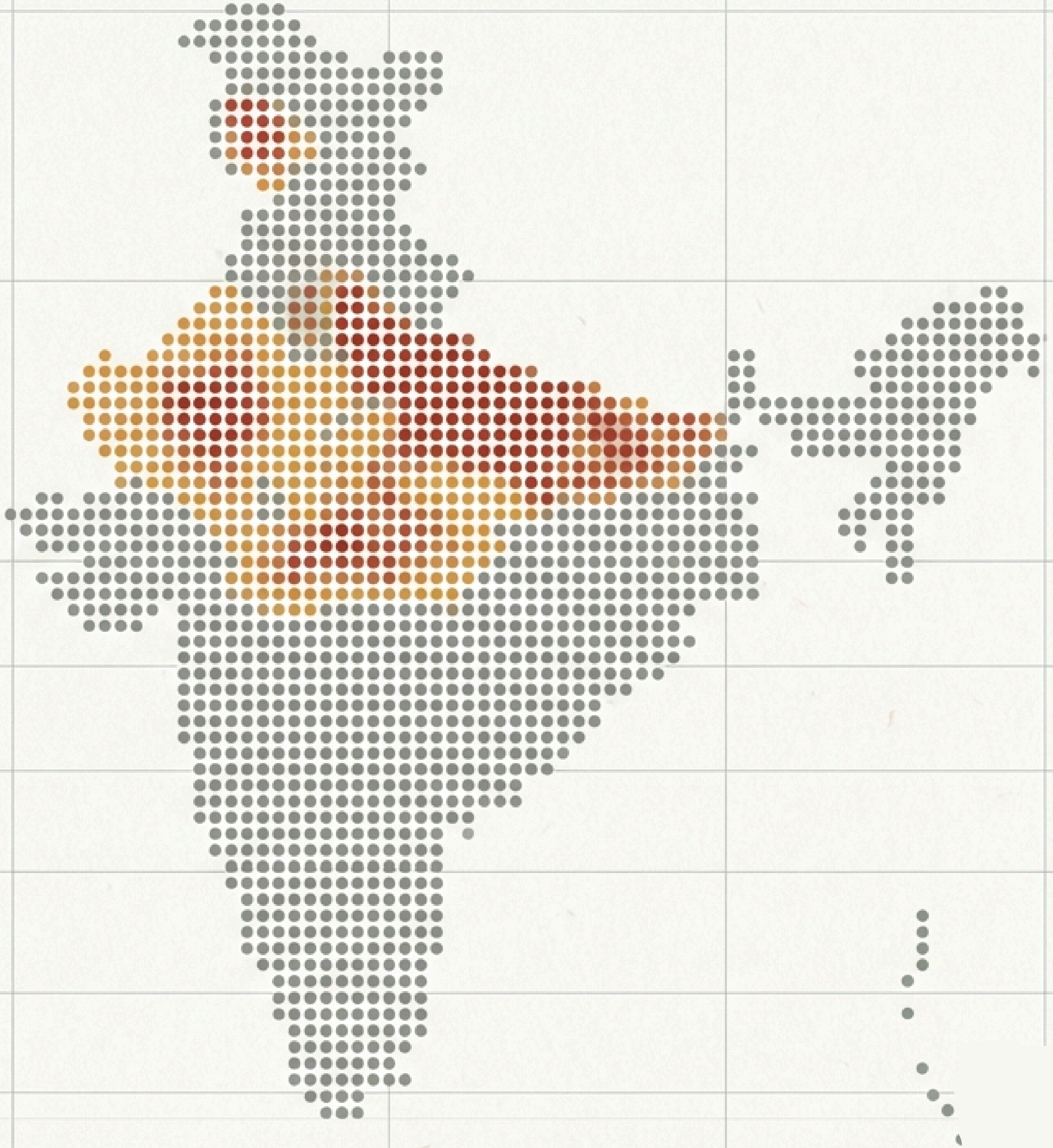
# The UIDAI Capacity Pressure Score (CPS) Framework

A State-Level Operational Stress Indicator for Aadhaar Update Operations

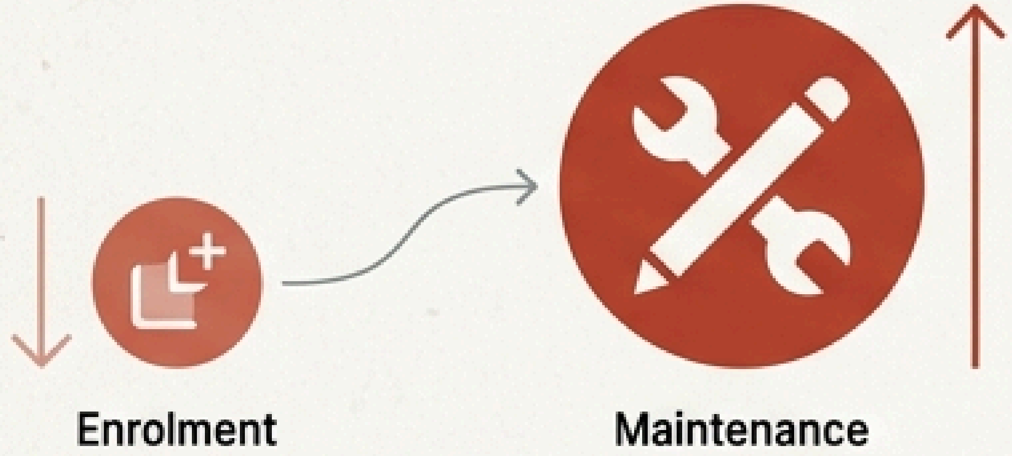


Author: Zaiba Muskan D.M
Context: UIDAI Aadhaar Data Hackathon
Code Repository: github team Id : UIDAI_3785

# As Aadhaar shifts from creation to maintenance, volume-based met’rics no longer capture true operational stress

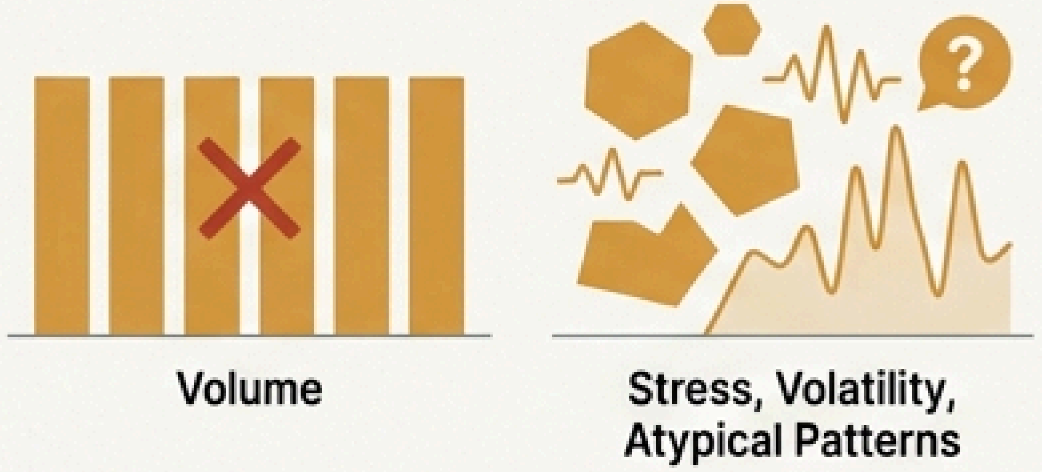
## The Context

Aadhaar has achieved near-universal enrolment. The system load has fundamentally shifted from new enrolments to maintenance-heavy update activity (demographic and biometric).



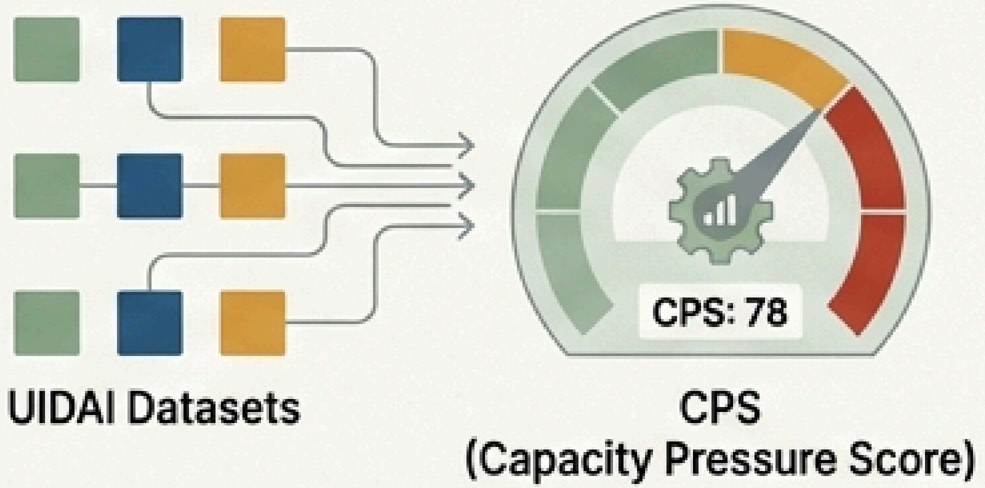

## The Problem

Traditional planning relies on raw volume. This fails to capture relative stress, volatility, and atypical update patterns.



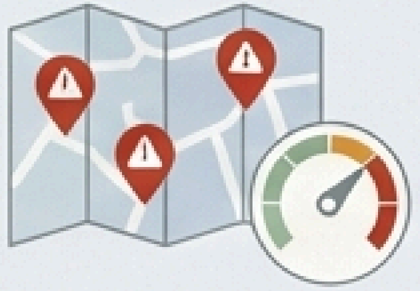

## The Solution

The Capacity Pressure Score (CPS)—a transparent, composite metric using existing UIDAI datasets to quantify operational pressure.



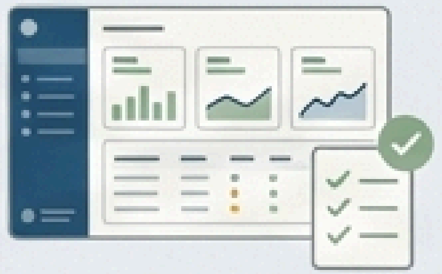

## Key Findings

**Identified:** 2 states under high sustained operational pressure (Red)



**Revealed:** Capacity stress is driven by update composition and volatility, not just volume

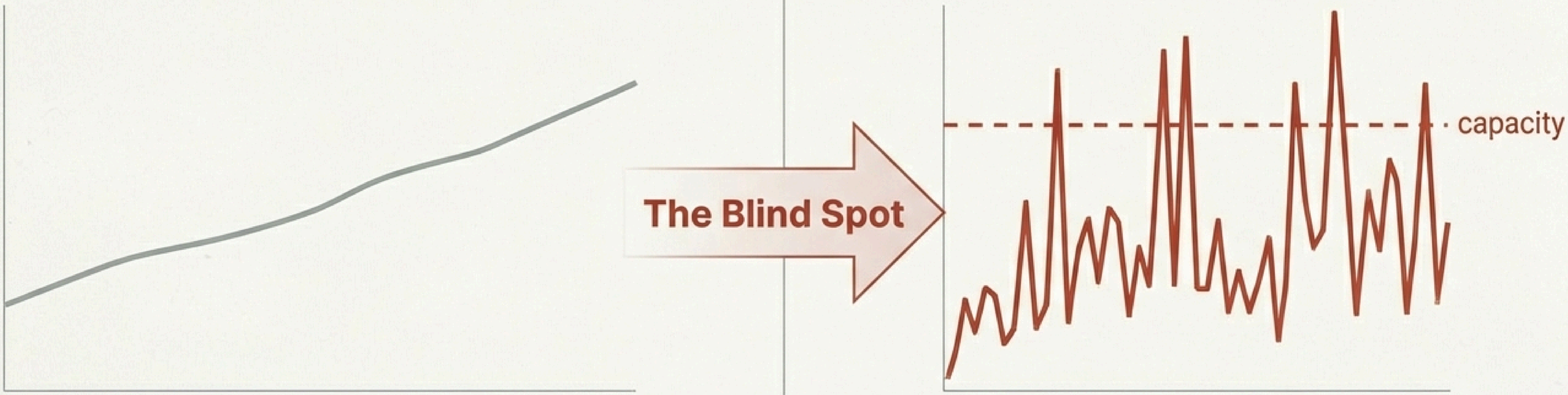

**Delivered:** A reproducible, auditable framework suitable for operational dashboards



# Volume is not Pressure: The blind spot in current capacity planning.

## The Old Metric: Volume

## The New Reality: Pressure




Counts how many people visit a centre. Good for scaling infrastructure during the enrolment phase. Fails to account for complexity.

Operations are now dominated by demographic edits and biometric updates. Without measuring relative stress, we risk service degradation while national averages look normal.

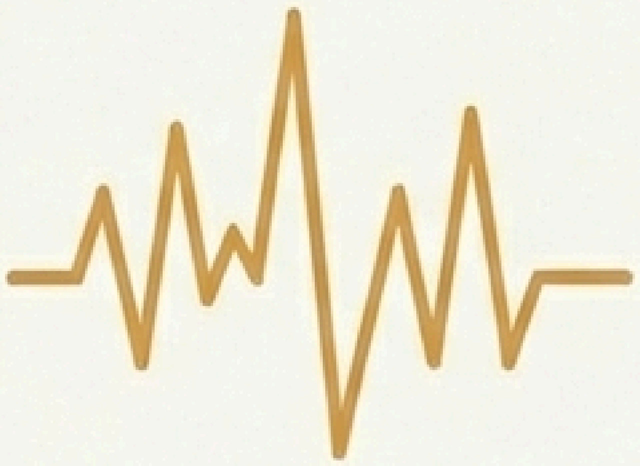
# Decomposing capacity stress into three interpretable signals.

### Signal 1: Update Intensity (Load)

Adult demographic updates relative to adult enrolments.

**Insight:** Captures the **Base Load**—sustained maintenance activity.

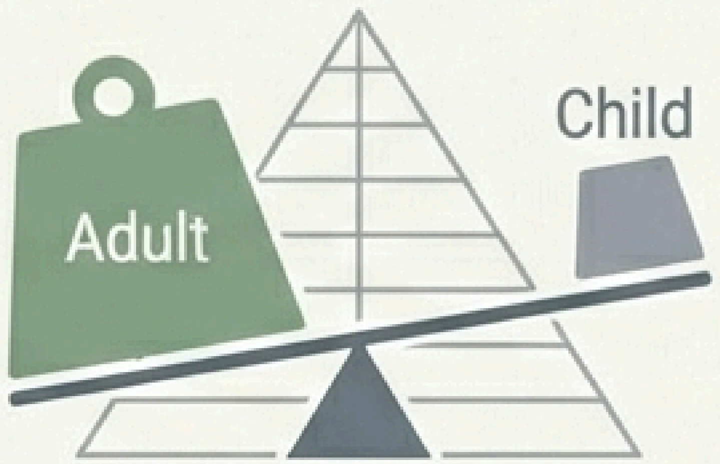
### Signal 2: Update Volatility (Instability)



Standard deviation of monthly update intensity.

**Insight:** Captures operational instability and surge behaviour. This is the **Stress Factor**.

### Signal 3: Adult–Child Skew (Complexity)




Relative dominance of adult updates over child updates.

**Insight:** Captures atypical update composition and demographic anomalies.

# The CPS Formula: A weighted composite prioritising sustained load.

$$CPS = 0.40(\text{Intensity}) + 0.30(\text{Volatility}) + 0.30(\text{Skew})$$

Weights:

Weights: 40% on Intensity ensures we prioritise sustained load. The 30/30 split on **Volatility** and **Skew** captures instability without letting outliers dominate.

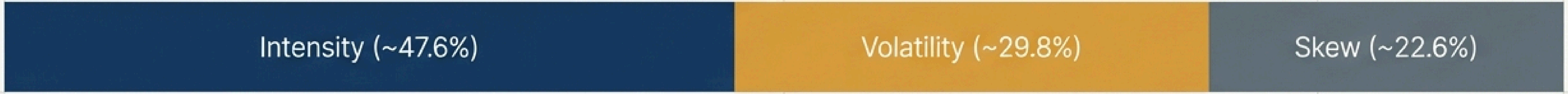
Normalisation:

All components are normalised to a 0–100 scale.

Nature of Score:

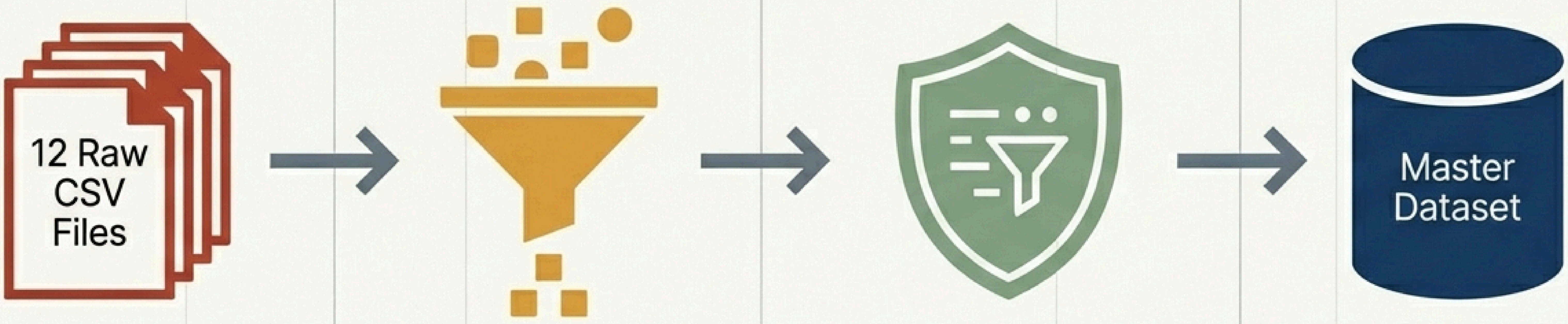
CPS is **comparative** (ranking states against each other), not absolute.

Observed Contribution to Final Score



# From 8.5 million events to a single auditable score.

| Input | Cleaning | Safeguards | Output |
|---|---|---|---|




**Input**

**Source:** UIDAI Aggregated Datasets (Enrolment, Demographic, Biometric).

**Scope:** 23 States, 141 State-Month observations.

**Cleaning**

Standardised inconsistent state naming (e.g., merging Andaman variants).
Removed malformed entries.

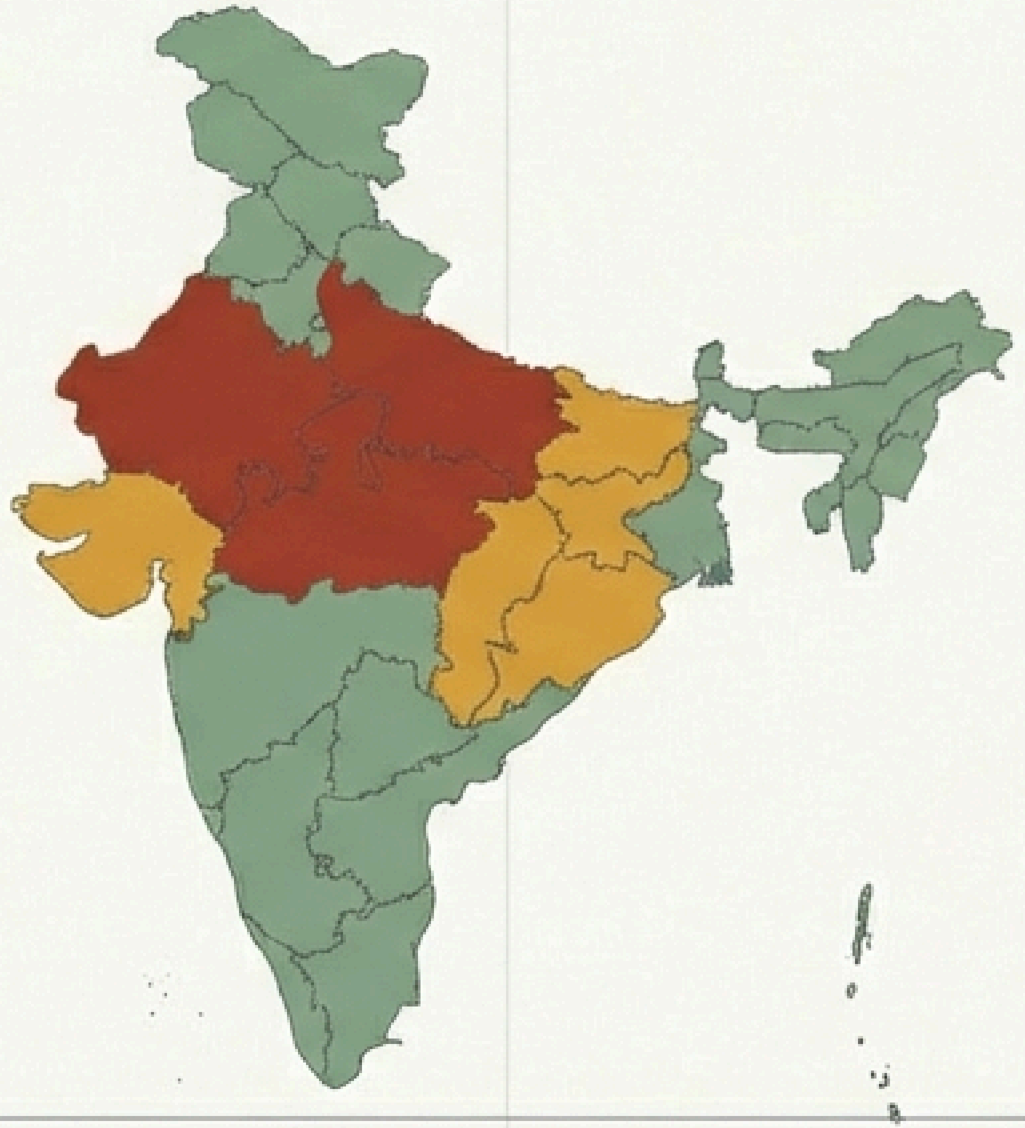
**Safeguards**

Excluded state-months with <100 adult enrolments to prevent ratio inflation in sparse regions.

**Output**

Clean, consistent observations for March–Dec 2025.

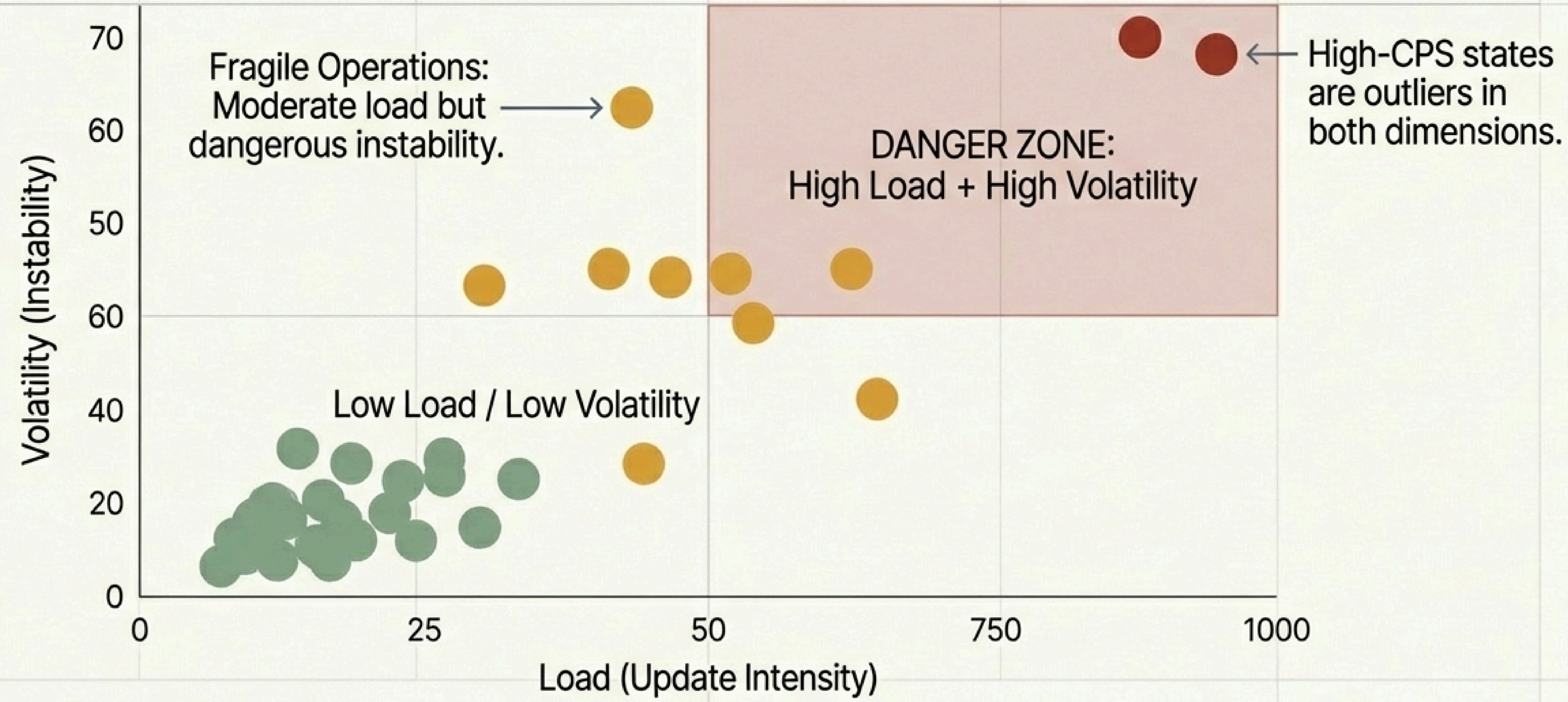
# The National Heatmap reveals stress is concentrated in specific regions.



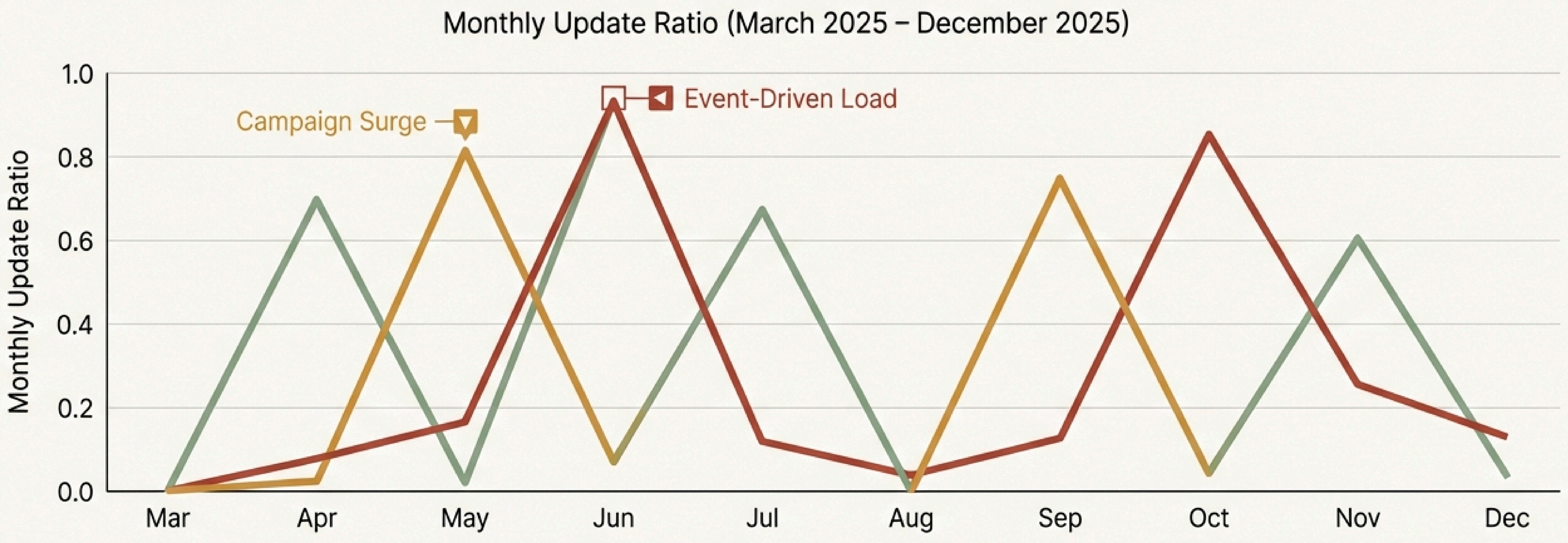
- Red (CPS ≥ 70): High pressure — immediate attention required (2 States)
- Yellow (40–69): Moderate pressure — monitoring needed (6 States)
- Green (< 40): Capacity broadly aligned with demand (15 States)

**Core Insight:** Several large states exhibit sustained operational stress, even when raw enrolment volumes appear moderate. This suggests hidden bottlenecks.

# High-pressure states cluster in zones of high load and high volatility.



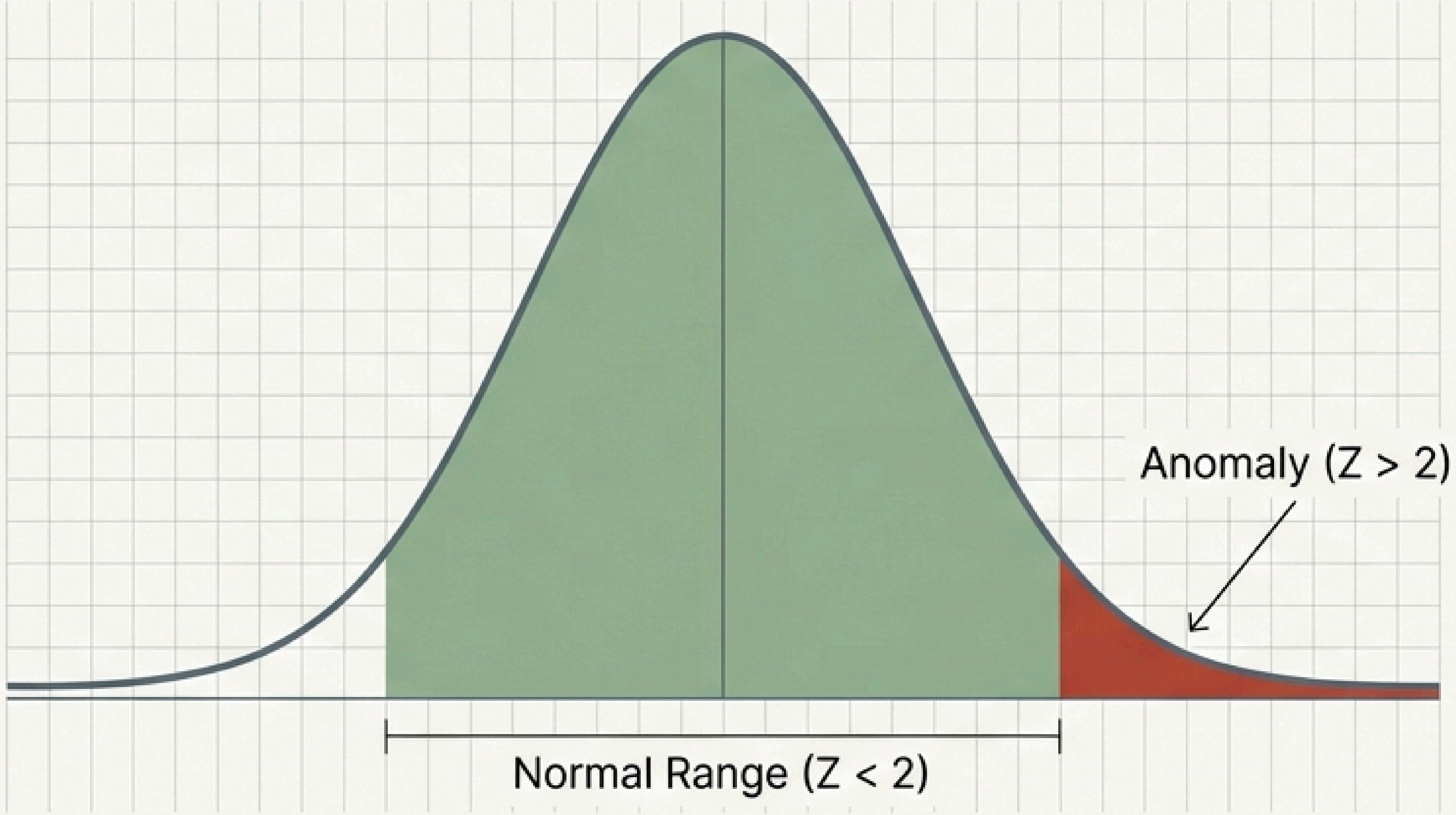
# Capacity stress is episodic, driven by campaign-like surges.




**Insight: Pressure** is dynamic. Sharp spikes indicate local government mandates or update campaigns rather than organic growth. Static resource allocation fails in this environment.

# Statistical anomaly detection flags sudden operational fractures.

## Z-Score methodology




Threshold: Two standard deviations from the mean.

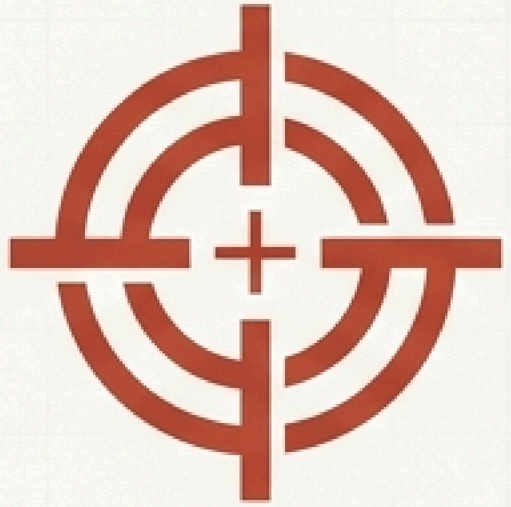
## Results Summary

**6**

Anomalous State-Months Identified

- Update ratios spiked abnormally in these specific months.
- Triggers a 'Diagnostic Review'.
- Identifies outliers that averages cannot capture.

# Transforming fragmented data into actionable operational levers



## Targeted Capacity Planning

Stop uniform national scaling. Focus resources (staff, kits) specifically on the high-pressure “Red” states.

## Early Warning System

Detect short-term surges in ‘Yellow’ states before they lead to service degradation or queues.

## Structural Diagnosis

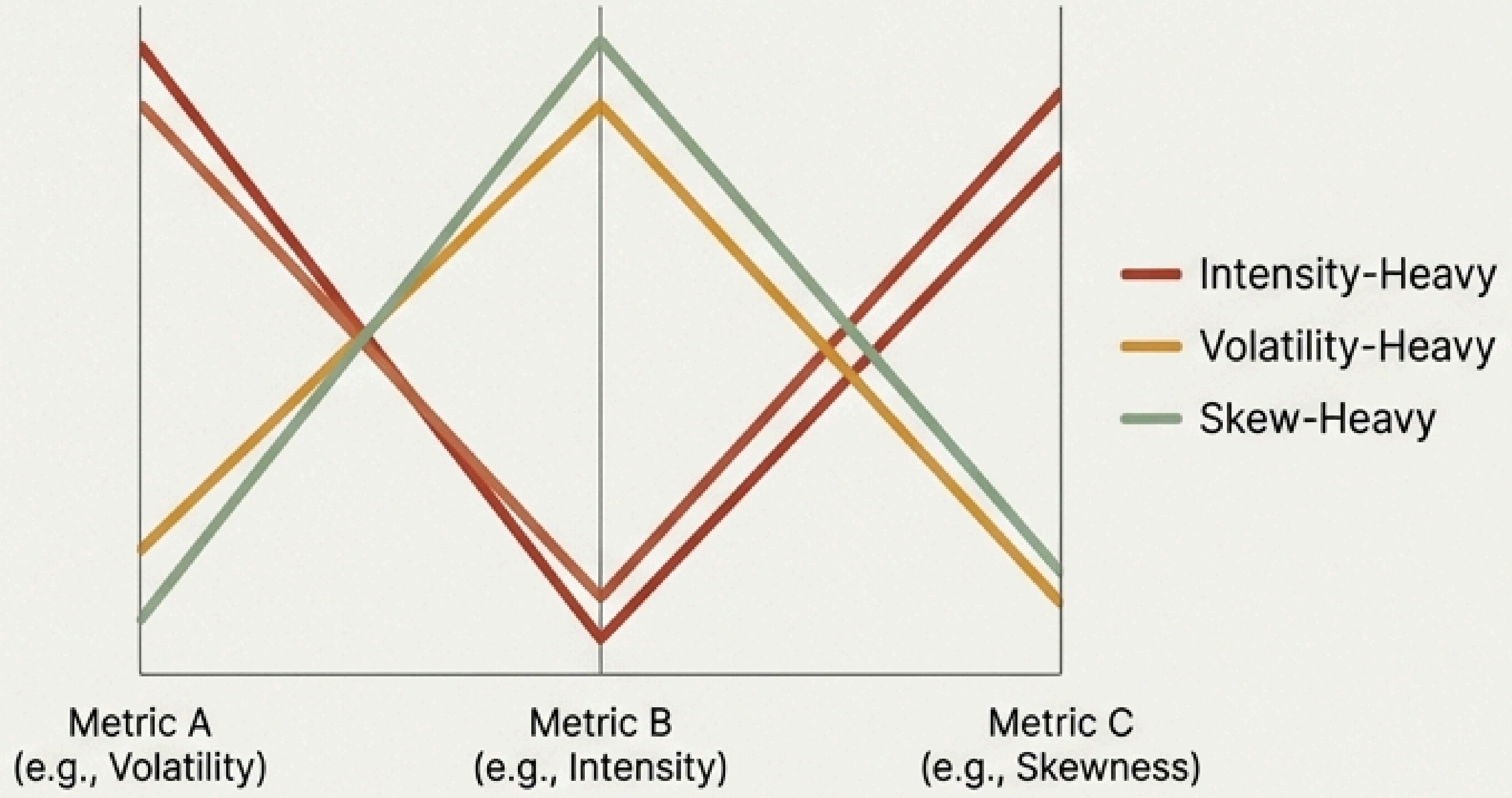
Distinguish between states with consistently high maintenance demand versus those suffering from episodic, campaign-driven stress.

## Evidence-Based Monitoring

Create internal dashboards that rely on auditable indicators rather than anecdotal reports.

# Robustness Checks: The signal remains consistent under stress testing.

## Sensitivity Analysis

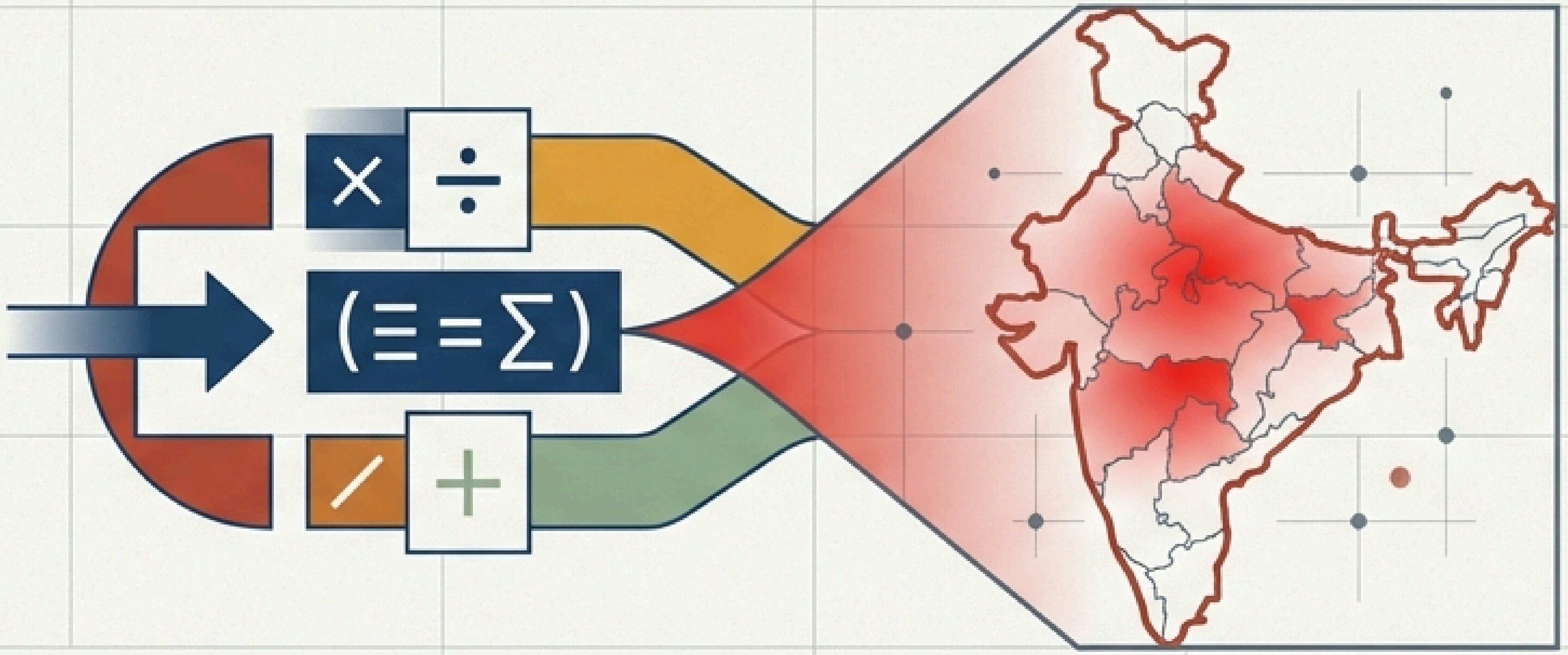



**Result:** The top high-pressure states remained consistent across all scenarios.

## Reliability & Limitations

- **Reconstruction Error:** 0.00 (Perfect reproducibility).
- **Limitation 1:** Relative Scores. Valid for state-to-state comparison, not historical.
- **Limitation 2:** Infrastructure Blind. Focuses on demand; supply-side data (staff/kits) to be added in future iterations.

# A transparent, reproducible lens for the next era of Aadhaar operations




From **reactive firefighting** based on raw volume → To **proactive management** using the Capacity Pressure Score

The CPS framework empowers **UIDAI** to **optimize** the ecosystem for a maintenance-heavy future, using only the data already in hand.